Machine Learning

Presented by Mr. Raviraj Solanki

Unit-1 Topics

- **What is human learning?**
- **What is Machine Learning?**
- **Human learning versus machine learning,**
- **Types of machine learning,**
- **Applications of machine learning,**
- **Tools for machine learning.**

Introduction of ML

- Machine learning is a growing technology which enables computers to learn automatically from past data.

- Machine learning uses various algorithms for **building mathematical models and making predictions using historical data or information**.

- Currently, it is being used for various tasks such as **image recognition**, **speech recognition**, **email filtering**, **Facebook auto-tagging**, **recommender system**, and many more.

Machine Learning Definitions

- **Algorithm:** A Machine Learning algorithm is a set of rules and statistical techniques used to learn patterns from data and draw significant information from it. It is the logic behind a Machine Learning model. An example of a Machine Learning algorithm is the **Linear Regression algorithm. (It is a statistical method that is used for predictive analysis.)**

- **Model:** A model is the main component of Machine Learning. **A model is trained by using a Machine Learning Algorithm**. An algorithm maps all the decisions that a model is supposed to take based on the given input, in order to get the correct output.

- **Predictor Variable:** It is a feature(s) of the data that can be used to predict the output.

- **Response Variable:** It is the feature or the output variable that needs to be predicted by using the predictor variable(s).

- **Training Data:** The Machine Learning **model is built using the training data.** The training data **helps the model to identify key trends and patterns essential to predict the output.**

- **Testing Data:** After the model is trained, it **must be tested to evaluate how accurately it can predict an outcome. This is done by the testing data set.**

What is Machine Learning?

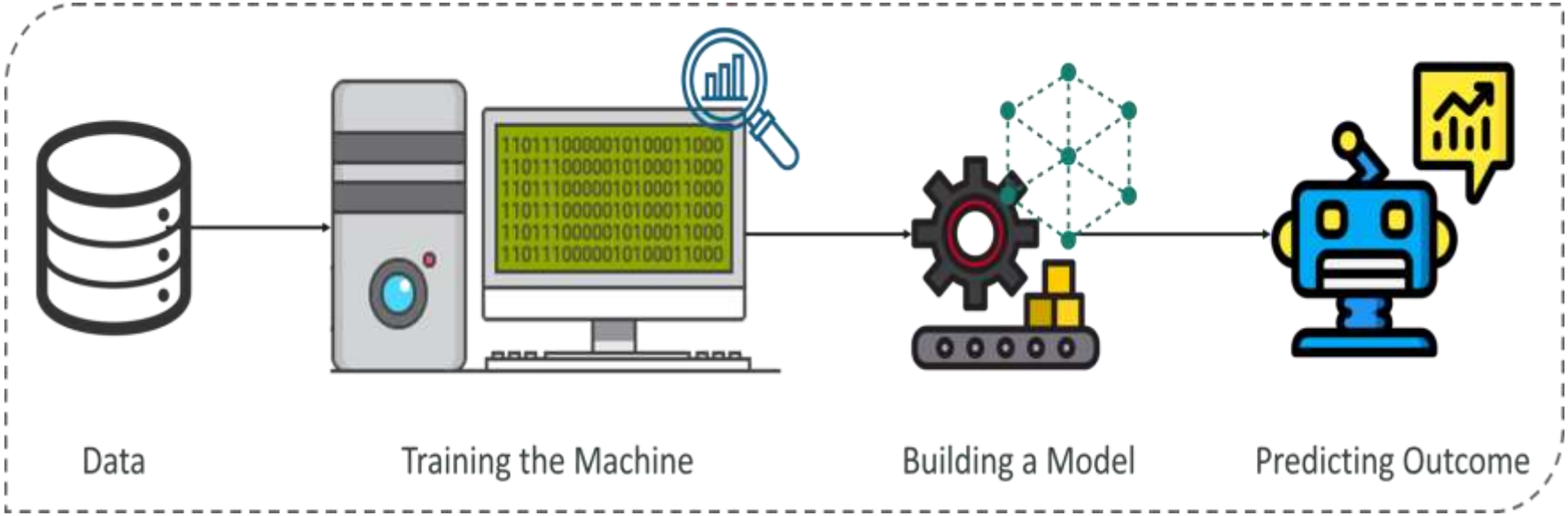
- Machine Learning is said as a subset of **artificial intelligence** that is mainly concerned with the **development of algorithms which allow a computer to learn from the data and past experiences on their own.**

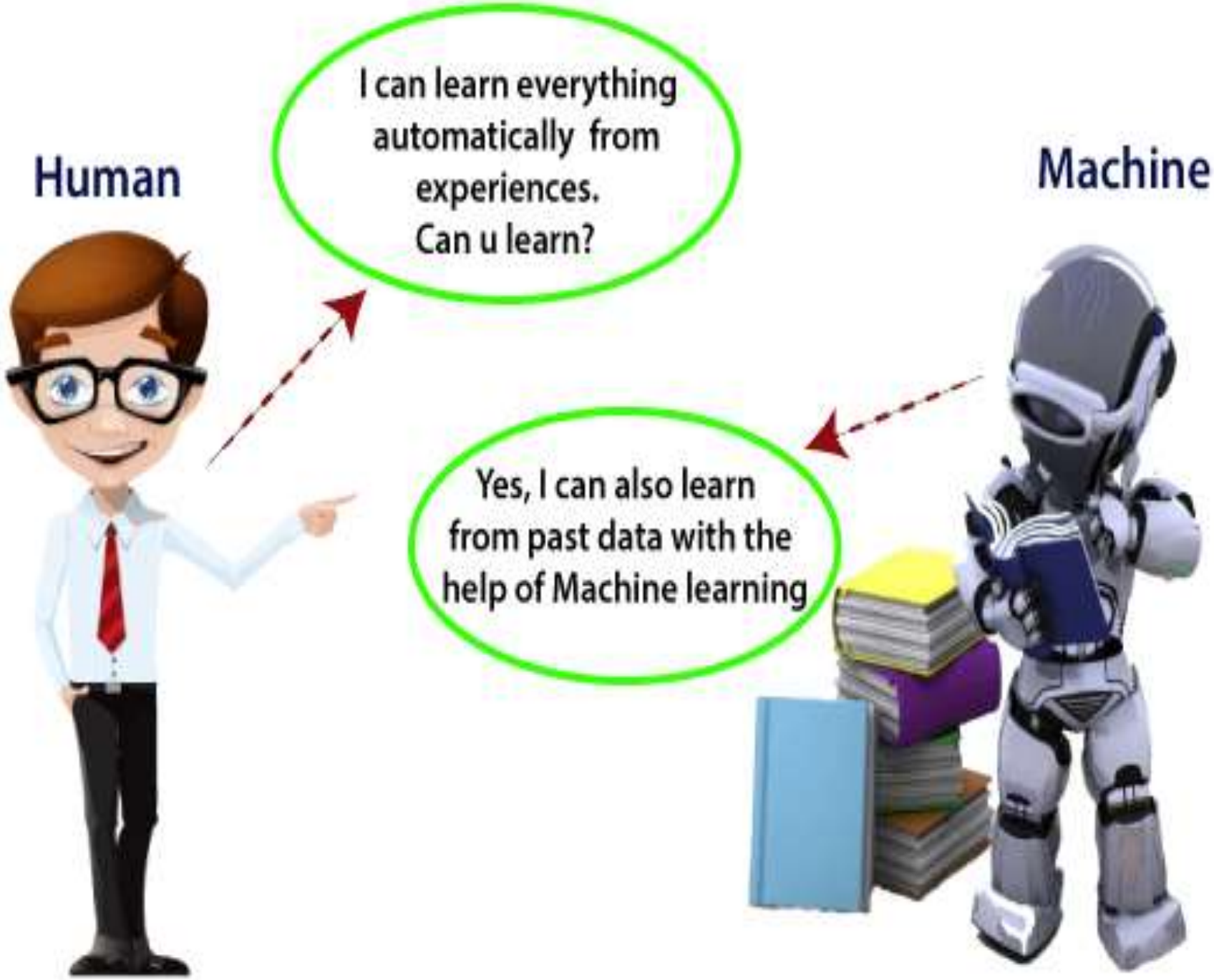
- The term machine learning was first introduced by **Arthur Samuel** in **1959**.

- In simple words, **ML is a type of artificial intelligence that extract patterns out of raw data by using an algorithm or method.**

- The **main focus of ML** is to allow computer systems learn from experience without being explicitly programmed or human intervention.

- Machine learning **enables a machine to automatically learn from data**, **improve performance from experiences**, and **predict things without being explicitly programmed**.


Data
Training the Machine
Building a Model
Predicting Outcome


Human
I can learn everything automatically from experiences. Can u learn?
Machine
Yes, I can also learn from past data with the help of Machine learning

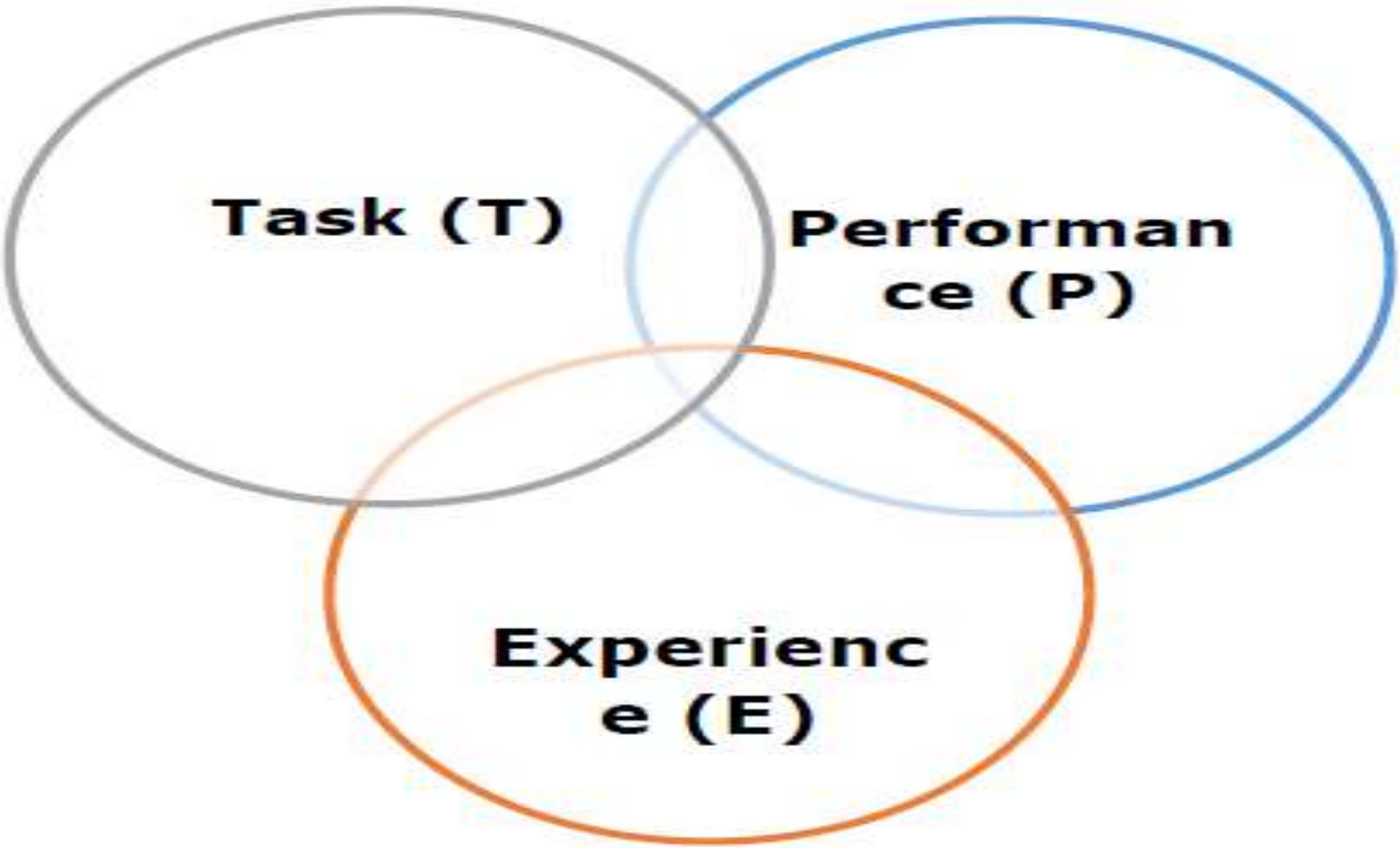
- With the help of sample historical data, which is known as **training data**, machine learning algorithms build a **mathematical model** that helps in making predictions or decisions without being explicitly programmed.

- Machine learning brings **computer science and statistics together** for **creating predictive models**.

- Machine learning constructs or **uses the algorithms** that learn from **historical data.**

- The **more we will provide the information**, the **higher will be the performance.**

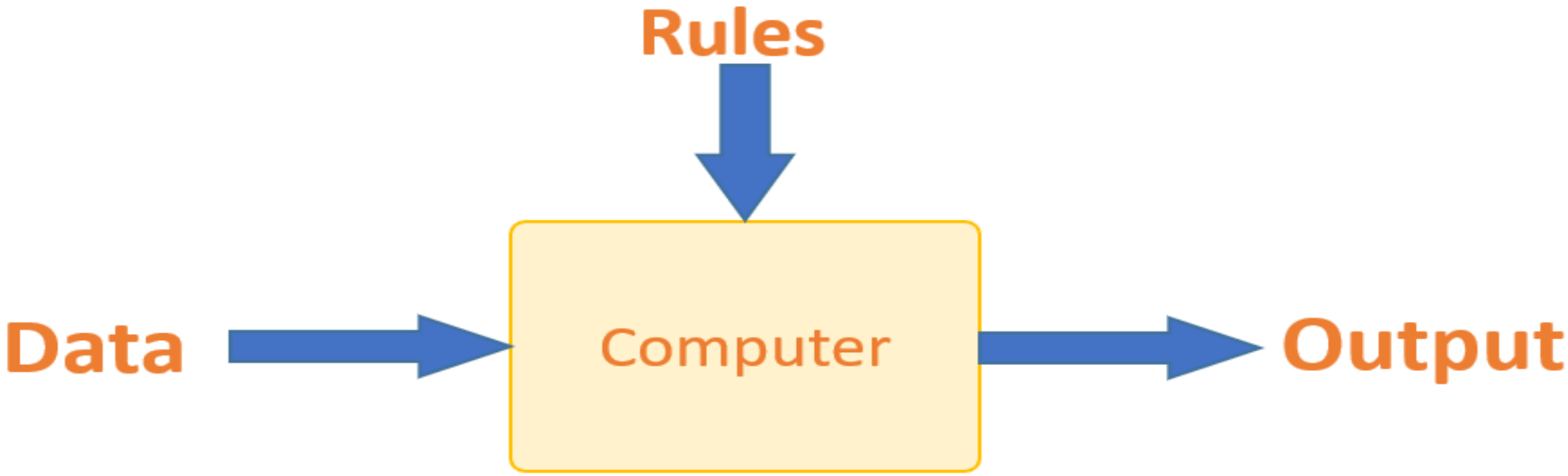
- **A machine has the ability to learn if it can improve its performance by gaining more data.**



Improve their performance (P)
At executing some task (T)
Over time with experience (E)

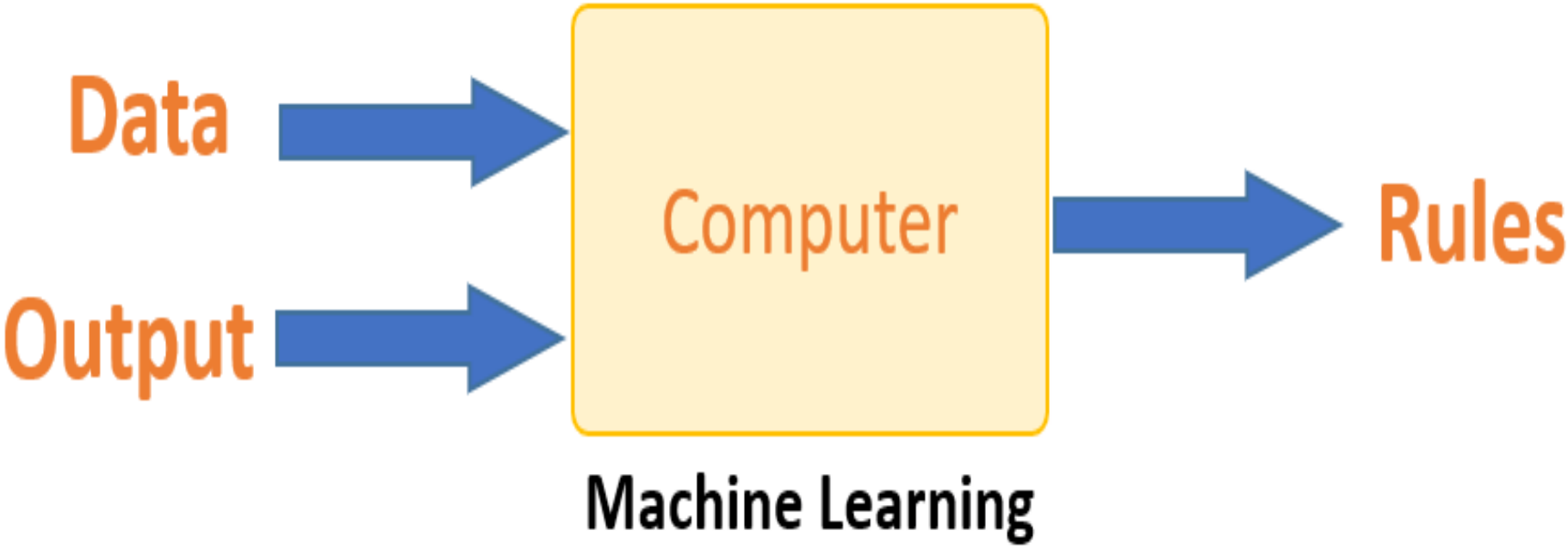
Human learning versus machine learning

- In traditional programming, a programmer code all the rules in consultation with an expert in the industry for which software is being developed.
- Each rule is based on a logical foundation; the **machine will execute an output following the logical statement.**
- **When the system grows complex**, **more rules** need to be written. It can **quickly become unsustainable** to maintain.



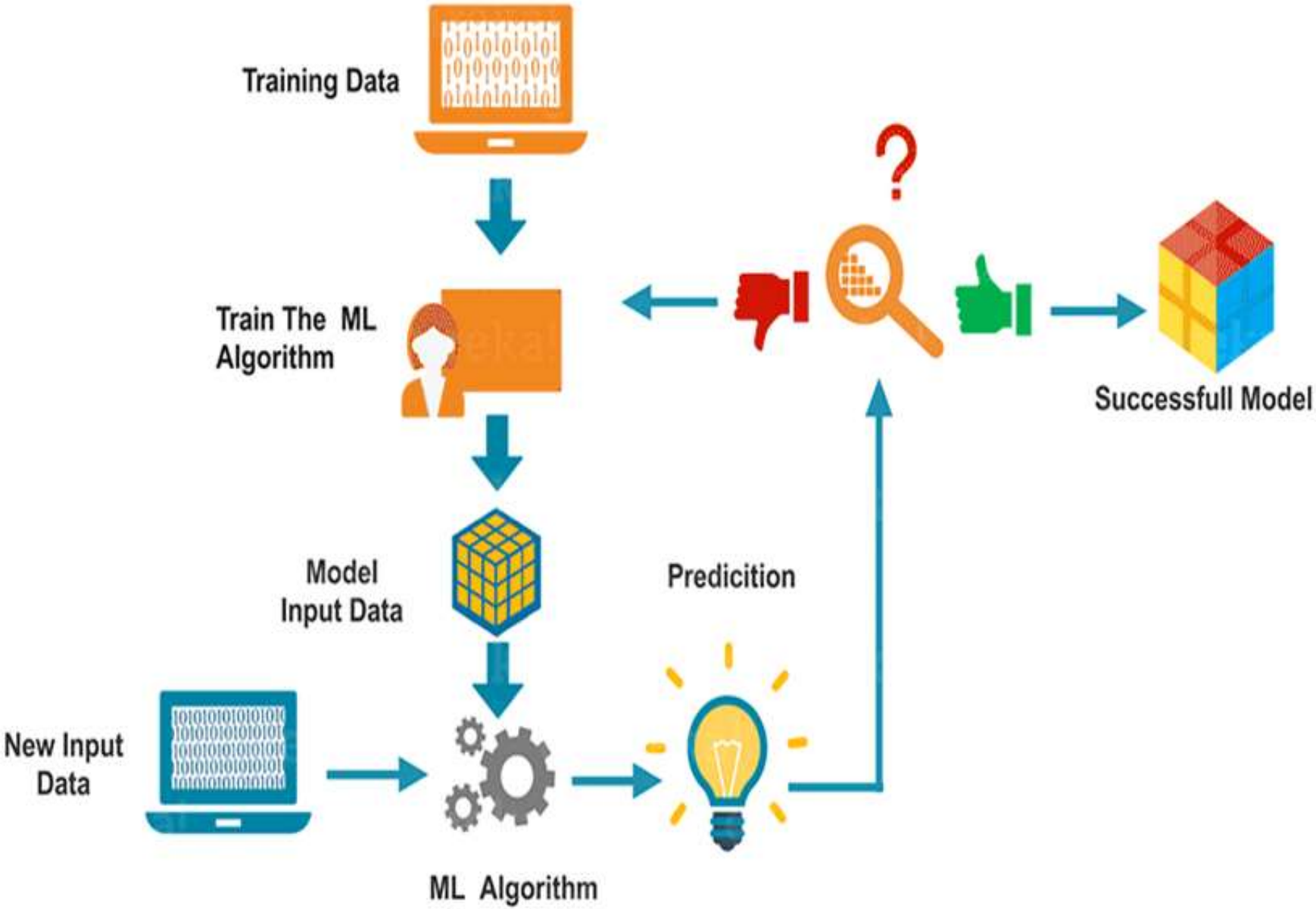
Traditional Programming

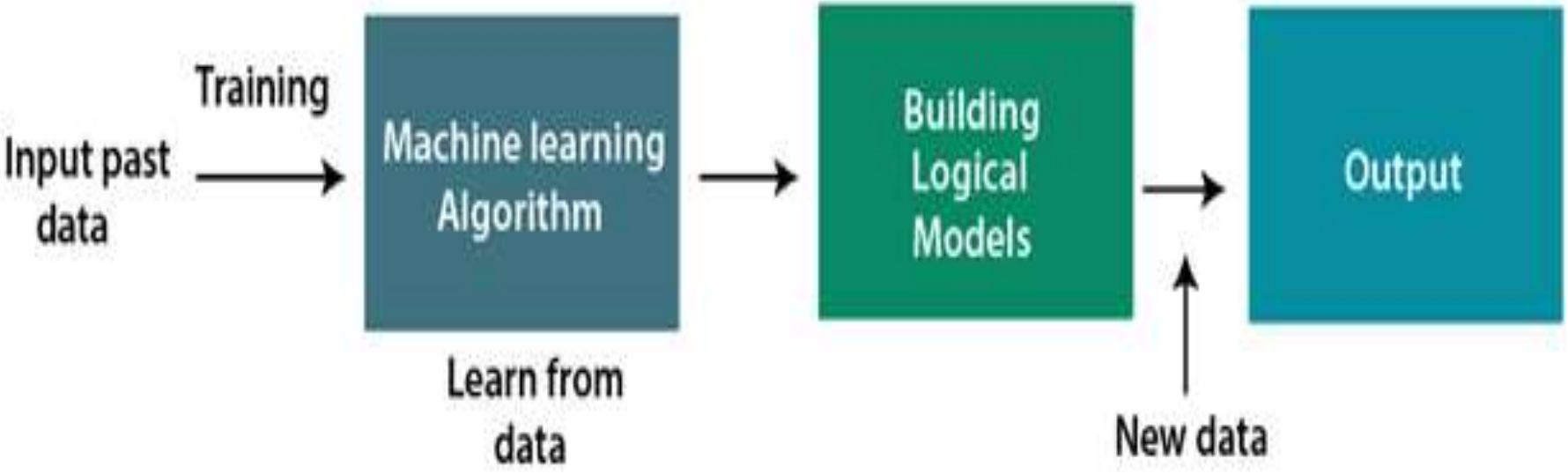
- Machine learning is supposed to overcome this issue.
- The machine learns how the input and **output data are correlated and it writes a rule.**
- The programmers do not need to write new rules each time there is new data.
- The algorithms adapt in response to new data and experiences to improve efficacy over time.



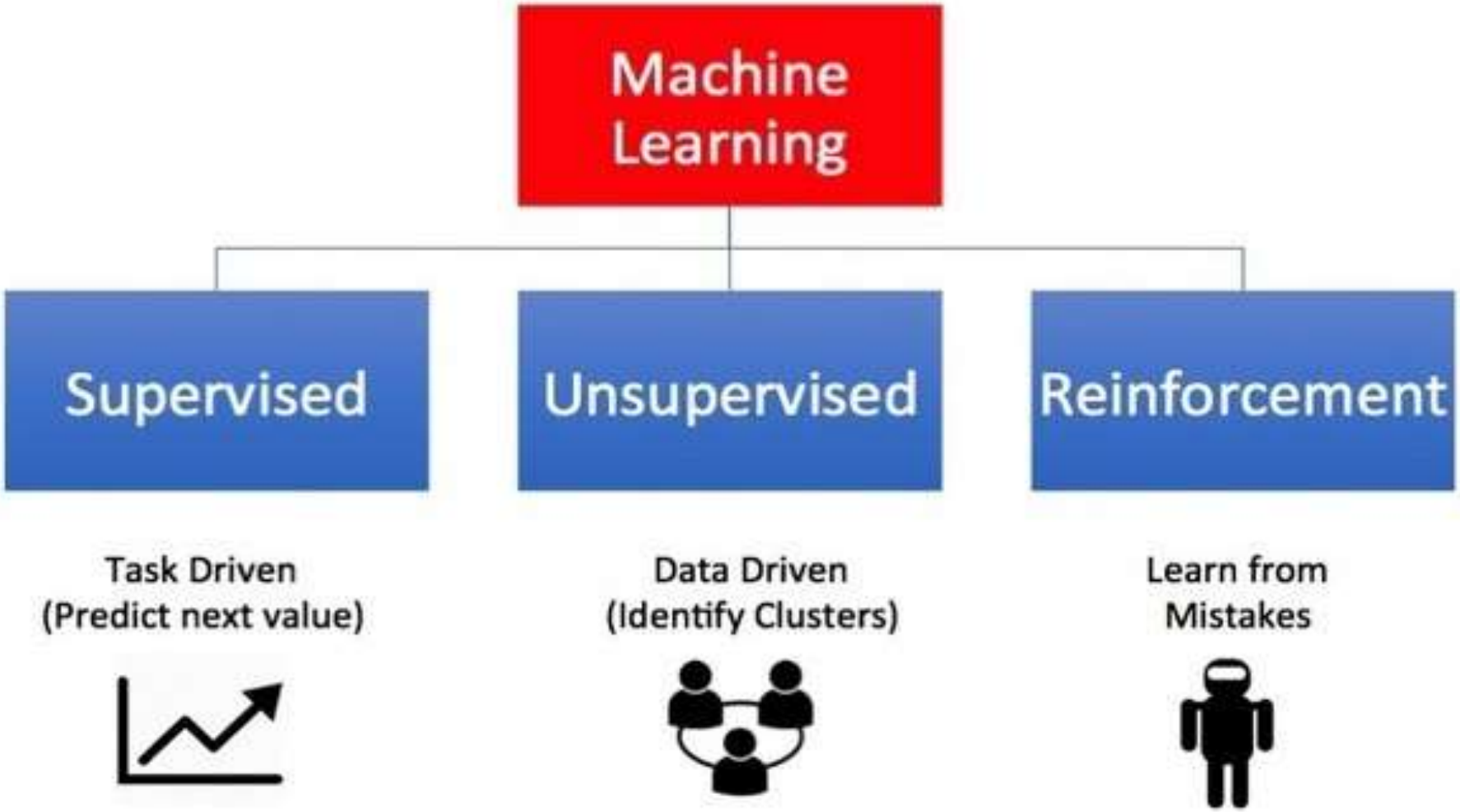
How does Machine Learning work

- A Machine Learning system **learns from historical data, builds the prediction models, and whenever it receives new data, predicts the output for it**.
- The accuracy of predicted output depends upon the amount of data, as the huge amount of data helps to build a better model which predicts the output more accurately.
- **Suppose we have a complex problem**, where we need to perform some predictions, so instead of writing a code for it, we just need to feed the data to generic algorithms, and **with the help of these algorithms, machine builds the logic as per the data and predict the output.**
- Machine learning has changed our way of thinking about the problem.


Training Data
Train The ML Algorithm
Successfull Model
Model Input Data
Predicition
New Input Data
ML Algorithm


Training
Input past data
Machine learning Algorithm
Learn from data
Building Logical Models
New data
Output

Types of machine learning



Types Of Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

Definition

Supervised learning is a method in which we teach the machine using labelled data

In unsupervised learning the machine is trained on unlabelled data without any guidance

In Reinforcement learning an agent interacts with its environment by producing actions & discovers errors or rewards

1) Supervised Learning

- Supervised learning is a type of machine learning method in **which we provide sample labeled data to the machine learning system in order to train it, and on that basis, it predicts the output.**

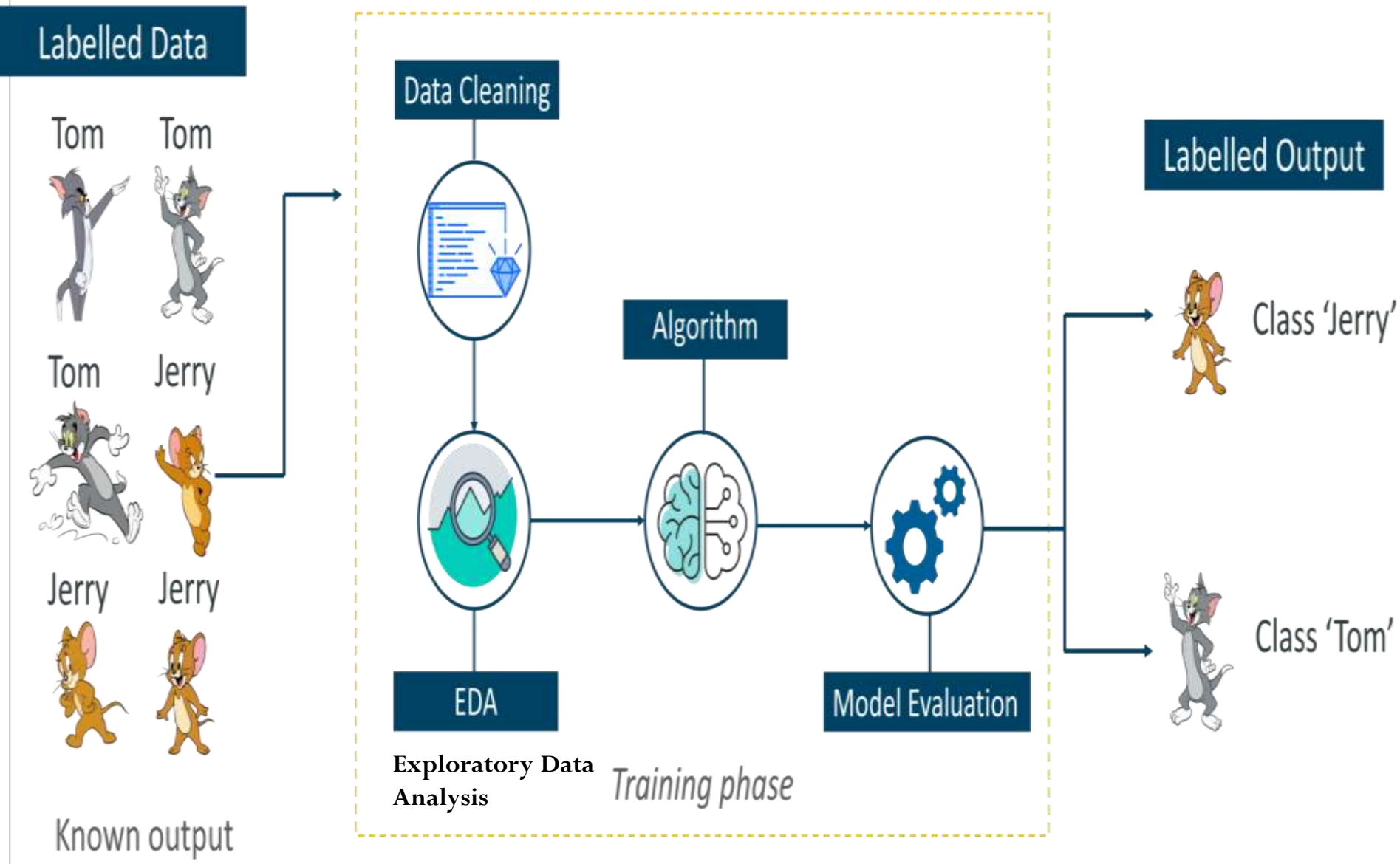
- **The labeled data set is nothing but the training data set.**

- The system creates a model using labeled data to understand the datasets and learn about each data, **once the training and processing are done then we test the model by providing a sample data to check whether it is predicting the exact output or not.**

- **The goal of supervised learning is to map input data with the output data.**

- The supervised learning is based on supervision, **and it is the same as when a student learns things in the supervision of the teacher.**

Example of Supervised Learning

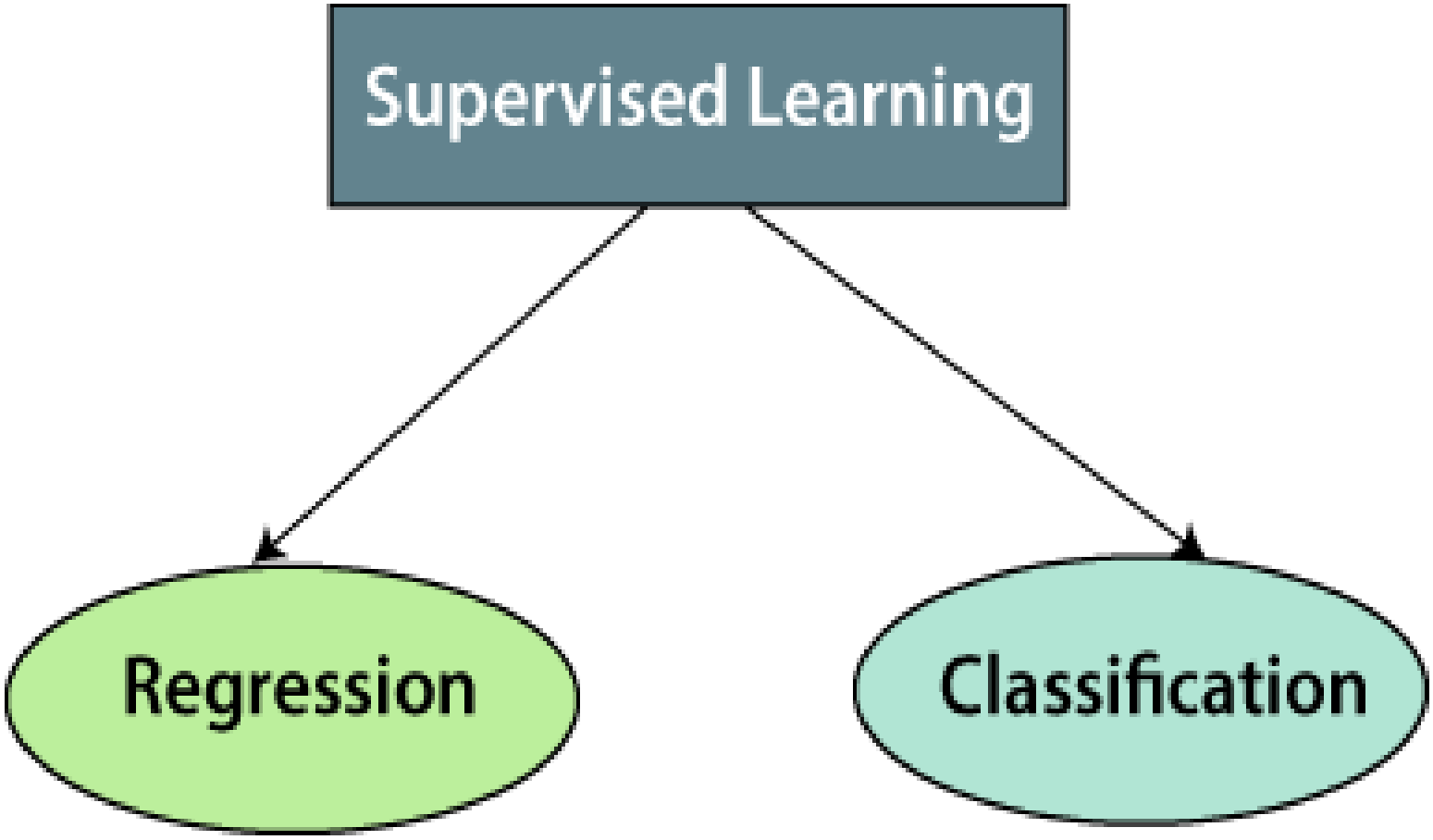


- The machine images of Tom and Jerry and the **goal is for the machine to identify and classify the images into two groups (Tom images and Jerry images).**

- The training data set that is fed to the model is labeled, as in, **we're telling the machine, 'this is how Tom looks and this is Jerry'.**

- By doing so you're training the machine by using labeled data. In Supervised Learning, there is a well-defined training phase done with the help of labeled data.

- Supervised learning can be grouped further in two categories of algorithms:



User ID	Gender	Age	Salary	Purchased
15624510	Male	19	19000	0
15810944	Male	35	20000	1
15668575	Female	26	43000	0
15603246	Female	27	57000	0
15804002	Male	19	76000	1
15728773	Male	27	58000	1
15598044	Female	27	84000	0
15694829	Female	32	150000	1
15600575	Male	25	33000	1
15727311	Female	35	65000	0
15570769	Female	26	80000	1
15606274	Female	26	52000	0
15746139	Male	20	86000	1
15704987	Male	32	18000	0
15628972	Male	18	82000	0
15697686	Male	29	80000	0
15733883	Male	47	25000	1

Figure A: CLASSIFICATION

- **Classification** : It is a Supervised Learning task where output is having defined labels(discrete value).

- **Classification algorithms are used when the output variable is categorical, which means there are two classes such as Yes-No, Male-Female, True-false, etc.**

- **For example in above Figure , Output – Purchased has defined labels i.e. 0 or 1 ;**

- **1 means the customer will purchase and 0 means that customer won't purchase.**

- **Classes can be called as target/labels or categories.**

- The output variable of classification is a category, not a value, such as “Green or blue”, “Fruit or animal”, etc.
- The goal here is to predict discrete values belonging to a particular class and evaluate on the basis of accuracy.
- It can be either binary or multi class classification.
- In **binary** classification, model predicts either 0 or 1 ; yes or no but in case of **multi class** classification, model predicts more than one class or more than two outcomes.
- **Example:** Gmail classifies mails in more than one classes like social, promotions, updates, forum.

- Below are some popular **classification algorithms** which come under supervised learning:

- **Random Forest**
- **Decision Trees**
- **Logistic Regression**
- **Support vector Machines**

- **Regression :** It is a Supervised Learning task where **output is having continuous value.**

- Regression algorithms are used if there is a relationship between the input variable and the output variable.

- It is **used for the prediction of continuous variables, such as Weather forecasting, Market Trends, etc.**

- Example in above Figure B, Output – Wind Speed is not having any discrete value but is continuous in the particular range.

- **The goal here is to predict a value as much closer to actual output value as our model can and then evaluation is done by calculating error value.**

- The smaller the error the greater the accuracy of our regression model.

Temperature	Pressure	Relative Humidity	Wind Direction	Wind Speed
10.69261758	986.882019	54.19337313	195.7150879	3.278597116
13.59184184	987.8729248	48.0648859	189.2951202	2.909167767
17.70494885	988.1119385	39.11965597	192.9273834	2.973036289
20.95430404	987.8500366	30.66273218	202.0752869	2.965289593
22.9278274	987.2833862	26.06723423	210.6589203	2.798230886
24.04233986	986.2907104	23.46918024	221.1188507	2.627005816
24.41475295	985.2338867	22.25082295	233.7911987	2.448749781
23.93361956	984.8914795	22.35178837	244.3504333	2.454271793
22.68800023	984.8461304	23.7538641	253.0864716	2.418341875
20.56425726	984.8380737	27.07867944	264.5071106	2.318677425
17.76400389	985.4262085	33.54900114	280.7827454	2.343950987
11.25680746	988.9386597	53.74139903	68.15406036	1.650191426
14.37810685	989.6819458	40.70884681	72.62069702	1.553469896
18.45114201	990.2960205	30.85038484	71.70604706	1.005017161
22.54895853	989.9562988	22.81738811	44.66042709	0.264133632
24.23155922	988.796875	19.74790765	318.3214111	0.329656571

Figure B: REGRESSION

- Below are some popular **Regression algorithms** which come under supervised learning:

- **Linear Regression**
- **Regression Trees**
- **Non-Linear Regression**
- **Bayesian Linear Regression**
- **Polynomial Regression**

Regression Algorithm	Classification Algorithm
In Regression, the output variable must be of continuous nature or real value.	In Classification, the output variable must be a discrete value.
The task of the regression algorithm is to map the input value (x) with the continuous output variable(y).	The task of the classification algorithm is to map the input value(x) with the discrete output variable(y).
Regression Algorithms are used with continuous data.	Classification Algorithms are used with discrete data.
In Regression, we try to find the best fit line, which can predict the output more accurately.	In Classification, we try to find the decision boundary, which can divide the dataset into different classes.
Regression algorithms can be used to solve the regression problems such as Weather Prediction, House price prediction, etc.	Classification Algorithms can be used to solve classification problems such as Identification of spam emails, Speech Recognition, Identification of cancer cells, etc.

Regression Algorithm	Classification Algorithm
The regression Algorithm can be further divided into Linear and Non-linear Regression.	The Classification algorithms can be divided into Binary Classifier and Multi-class Classifier.

Advantages of Supervised learning:

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- In supervised learning, we can have an exact idea about the classes of objects.
- Supervised learning model helps us to solve various real-world problems such as **fraud detection, spam filtering**, etc.

Disadvantages of supervised learning:

- Supervised learning models are **not suitable for handling the complex tasks.**
- Supervised learning **cannot predict the correct output if the test data is different from the training dataset.**
- Training required **lots of computation times**.
- In supervised learning, we need enough knowledge about the classes of object.

2) Unsupervised Learning

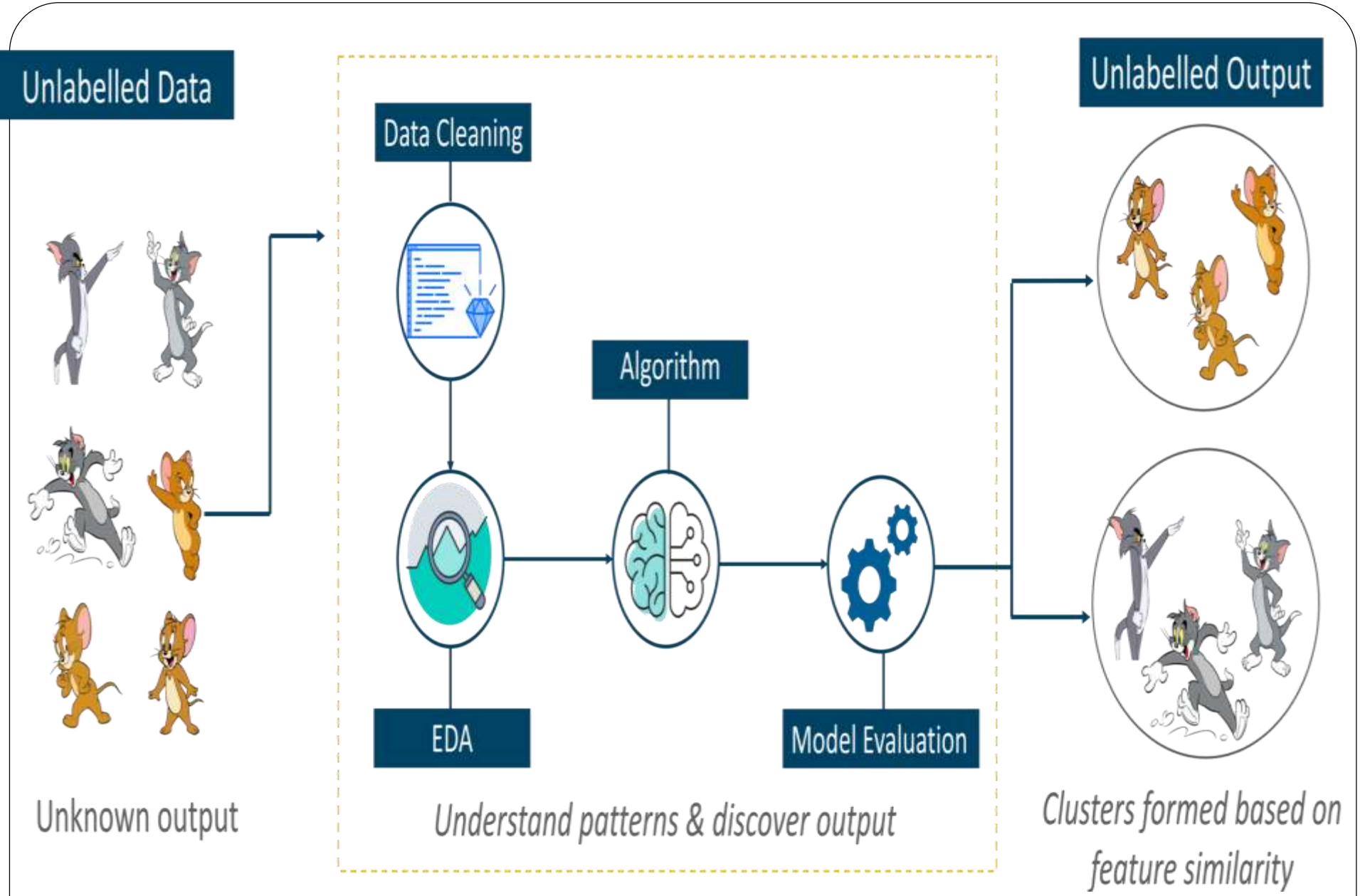
- Unsupervised learning is a learning method **in which a machine learns without any supervision.**

- The training is provided to **the machine with the set of data that has not been labeled, classified, or categorized, and the algorithm needs to act on that data without any supervision**.

- The **goal of unsupervised learning is to restructure the input data into new features or a group of objects with similar patterns.**

- In unsupervised learning, we don't have a predetermined result. The machine tries to find useful insights from the huge amount of data.

- It can be further classifieds into two categories of algorithms:
- **Clustering**
- **Association**



EDA : Exploratory data analysis

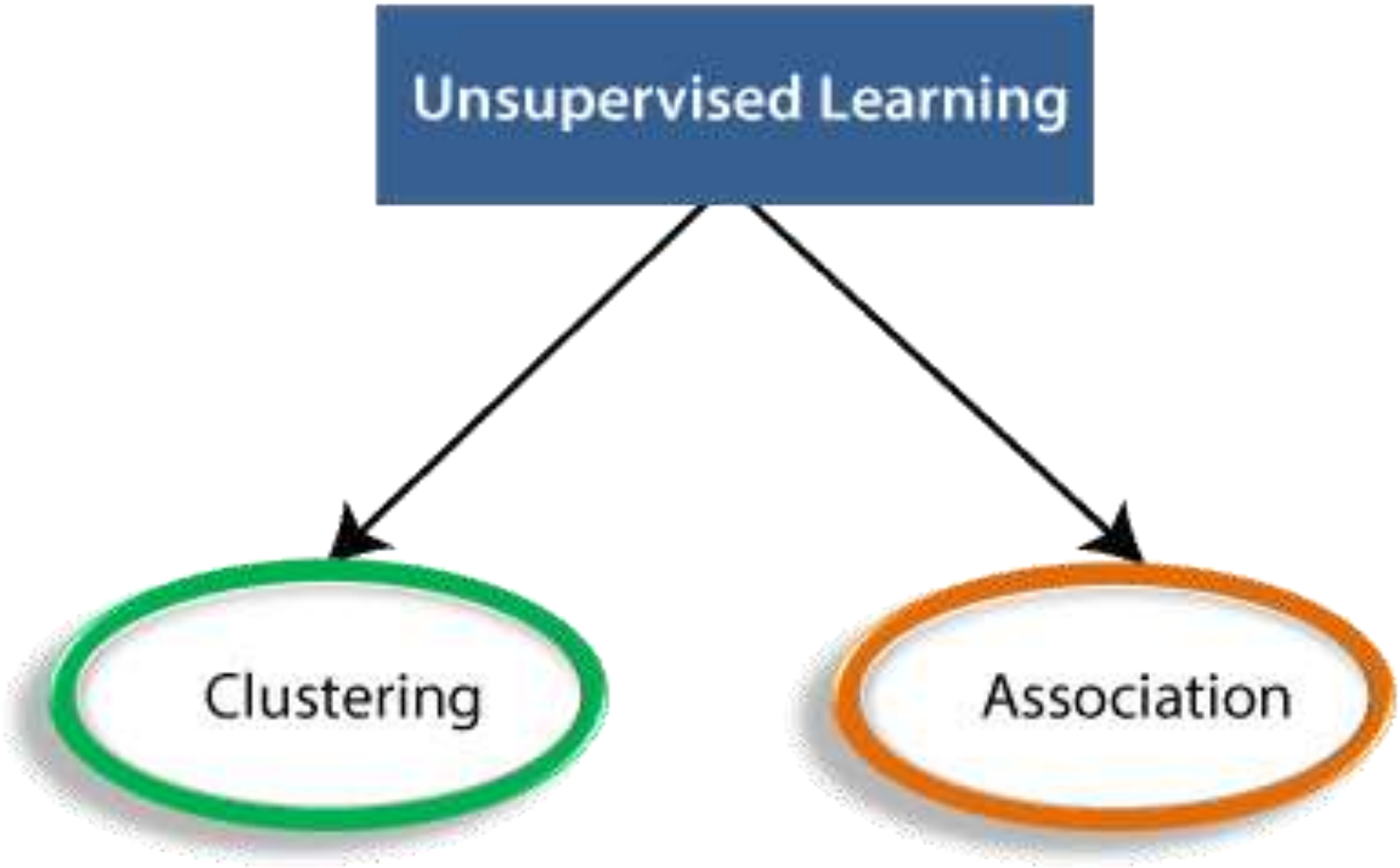
- **For example**, it identifies prominent features of **Tom such as pointy ears, bigger size, etc, to understand that this image is of type 1.**

- Similarly, it finds such features in Jerry and knows that this image is of type 2.

- Therefore, it classifies the images into two different classes without knowing who Tom is or Jerry is.

Why Unsupervised Learning?

- Unsupervised machine learning **finds all kind of unknown patterns in data.**
- Unsupervised methods **help you to find features which can be useful for categorization.**
- It is taken place in real time, so all the input data to be analyzed and labeled in the presence of learners.
- It is **easier to get unlabeled data from a computer than labeled data, which needs manual intervention.**


Unsupervised Learning
Clustering
Association

- **Clustering**: Clustering is a **method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group.**

- Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.

- **Association:** An association rule is an unsupervised learning method which is used for **finding the relationships between variables in the large database.**
- It determines the set of items that occurs together in the dataset.
- Association rule makes marketing strategy more effective.
- Such as people who buy X item (suppose a bread) are also tend to purchase Y (Butter/Jam) item.
- A typical example of **Association rule is Market Basket Analysis.**

Unsupervised Learning algorithms:

- **K-means clustering**
- **KNN (k-nearest neighbors)**
- **Hierarchal clustering**
- **Anomaly detection**
- **Neural Networks**
- **Principle Component Analysis**
- **Independent Component Analysis**
- **Apriori algorithm**
- **Singular value decomposition**

Advantages of Unsupervised Learning

- Unsupervised learning is used for **more complex tasks** as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.

- Unsupervised learning is preferable as **it is easy to get unlabeled data** in comparison to labeled data.

Disadvantages of Unsupervised Learning

- Unsupervised learning is basically more difficult than supervised learning as it **does not have corresponding output.**

- The result of the unsupervised learning algorithm might **be less accurate as input data is not labeled**, and **algorithms do not know the exact output in advance.**

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.

Supervised Learning	Unsupervised Learning
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.
Supervised learning can be categorized in Classification and Regression problems .	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.

Supervised Learning	Unsupervised Learning
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.

3) Reinforcement Learning

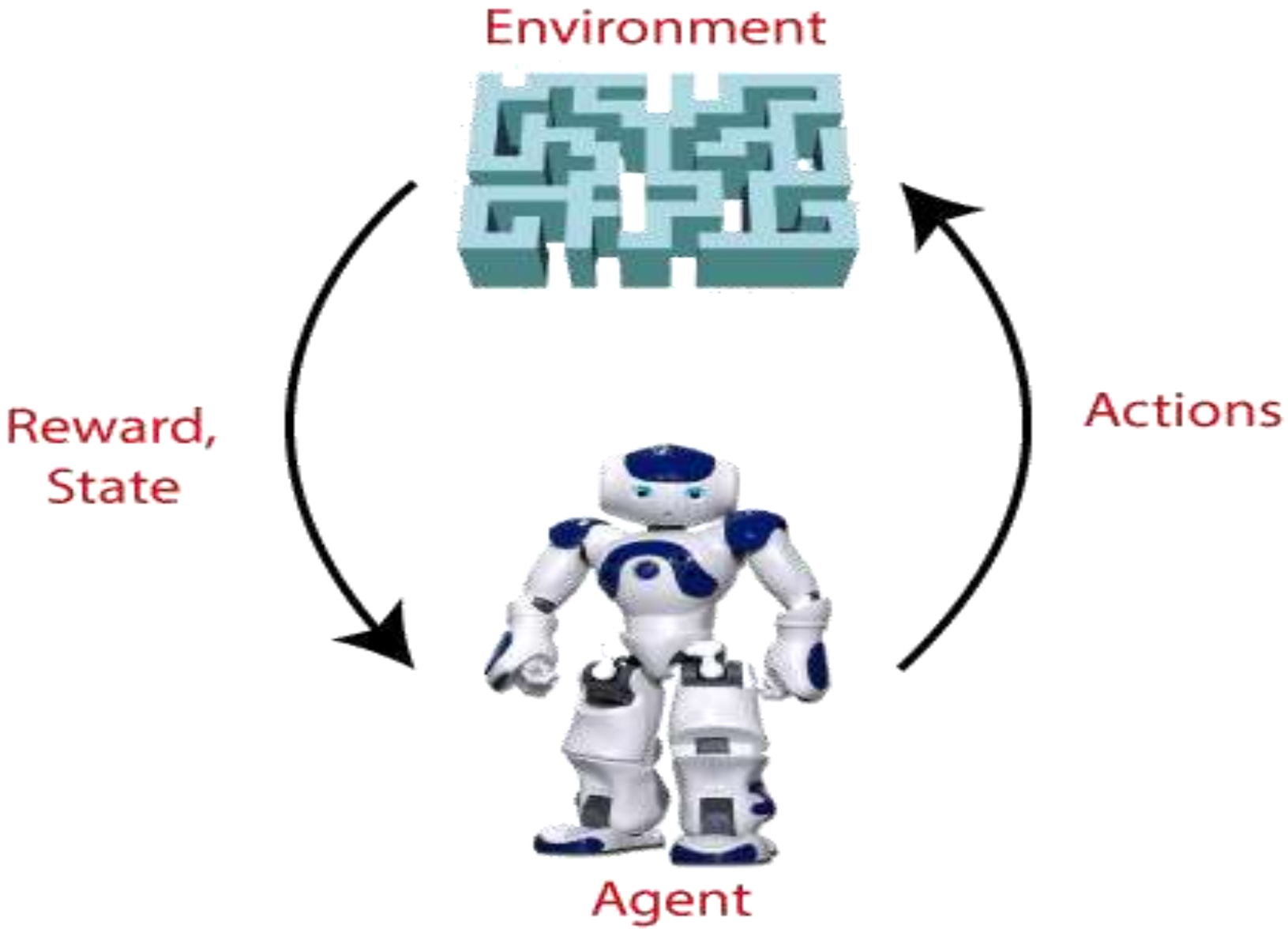
- Reinforcement learning is **a feedback-based learning method**, in which a **learning agent gets a reward for each right action** and **gets a penalty for each wrong action.**

- The agent **learns automatically with these feedbacks and improves its performance.**

- In reinforcement learning, **the agent interacts with the environment and explores it.**

- The **goal of an agent is to get the most reward points, and hence, it improves its performance.**

- Reinforcement Learning is a part of Machine learning **where an agent is put in an environment and he learns to behave in this environment by performing certain actions and observing the rewards which it gets from those actions.**


Environment
Reward,
State
Actions
Agent

- **For ex**. You will learn how to live on the island. You will explore the environment, understand the climate condition, the type of food that grows there, the dangers of the island, etc. This is exactly how Reinforcement Learning works, it involves an Agent (you, stuck on the island) that is put in an unknown environment (island), where he must learn by observing and performing actions that result in rewards.

- Reinforcement Learning is **mainly used in advanced Machine Learning areas such as self-driving cars, AplhaGo, etc.**

- The robotic dog, which automatically learns the movement of his arms, is an example of Reinforcement learning.

some important terms used in Reinforcement

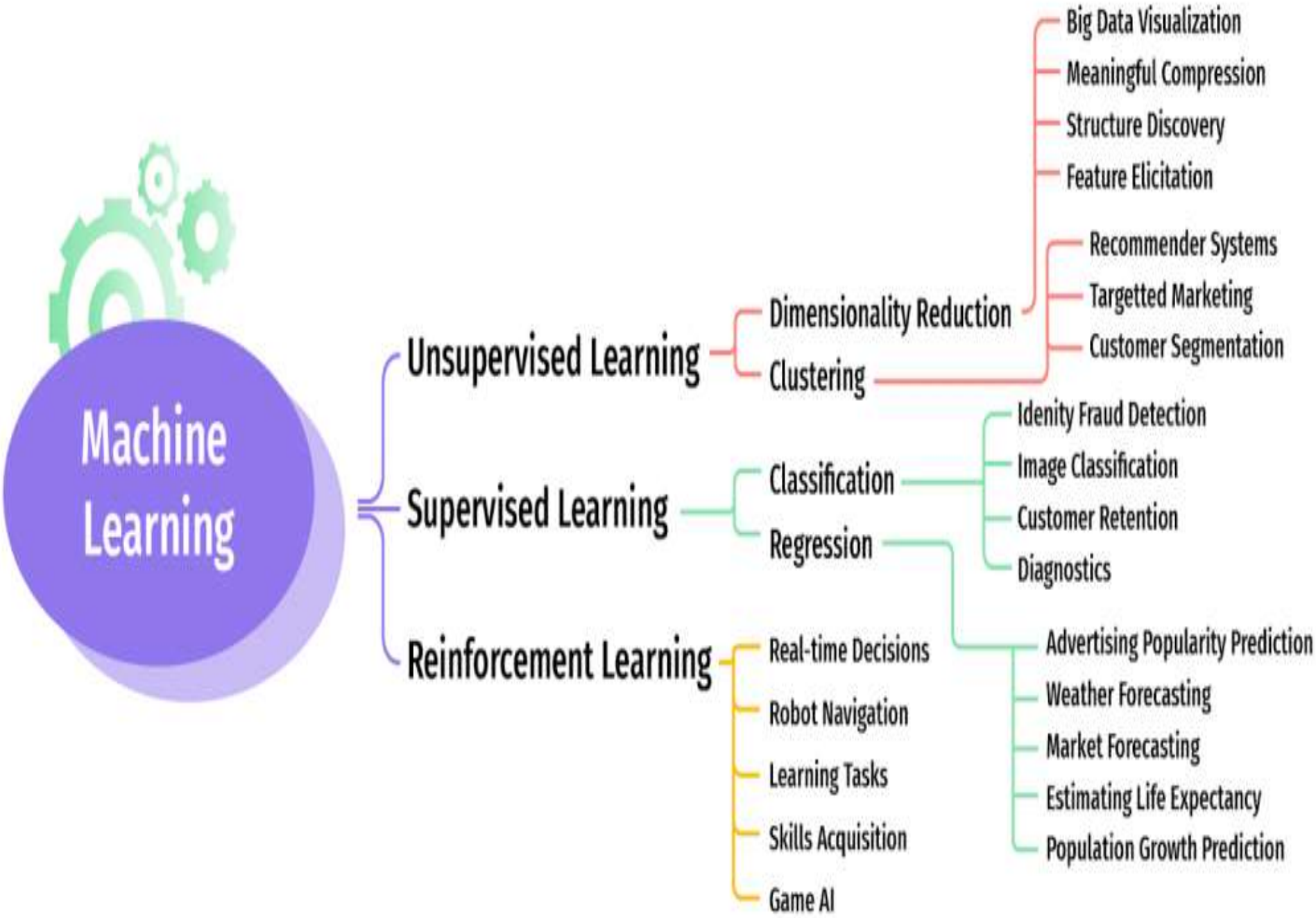
- **Agent:** It is an assumed entity which performs actions in an environment to gain some reward.
- **Environment (e):** A scenario that an agent has to face.
- **Reward (R):** An immediate return given to an agent when he or she performs specific action or task.
- **State (s):** State refers to the current situation returned by the environment.
- **Policy (π):** It is a strategy which applies by the agent to decide the next action based on the current state.
- **Value (V):** It is expected long-term return with discount, as compared to the short-term reward.
- **Value Function:** It specifies the value of a state that is the total amount of reward. It is an agent which should be expected beginning from that state.
- **Model of the environment:** This mimics the behavior of the environment. It helps you to make inferences to be made and also determine how the environment will behave.
- **Model based methods:** It is a method for solving reinforcement learning problems which use model-based methods.
- **Q value or action value (Q):** Q value is quite similar to value. The only difference between the two is that it takes an additional parameter as a current action.

Reinforcement Learning Algorithms

- **Value-Based:**
- In a value-based Reinforcement Learning method, you should try to maximize a value function **V(s)**. In this method, the agent is expecting a long-term return of the current states under policy π.
- **Policy-based:**
- In a policy-based RL method, you try to come up with such a policy that the action performed in every state helps you to gain maximum reward in the future.
- **Two types of policy-based methods are:**
- **Deterministic:** For any state, the same action is produced by the policy π.
- **Stochastic:** Every action has a certain probability, which is determined by the following equation.Stochastic Policy:
- **n{a\s) = P\A, = a\S, =S]**

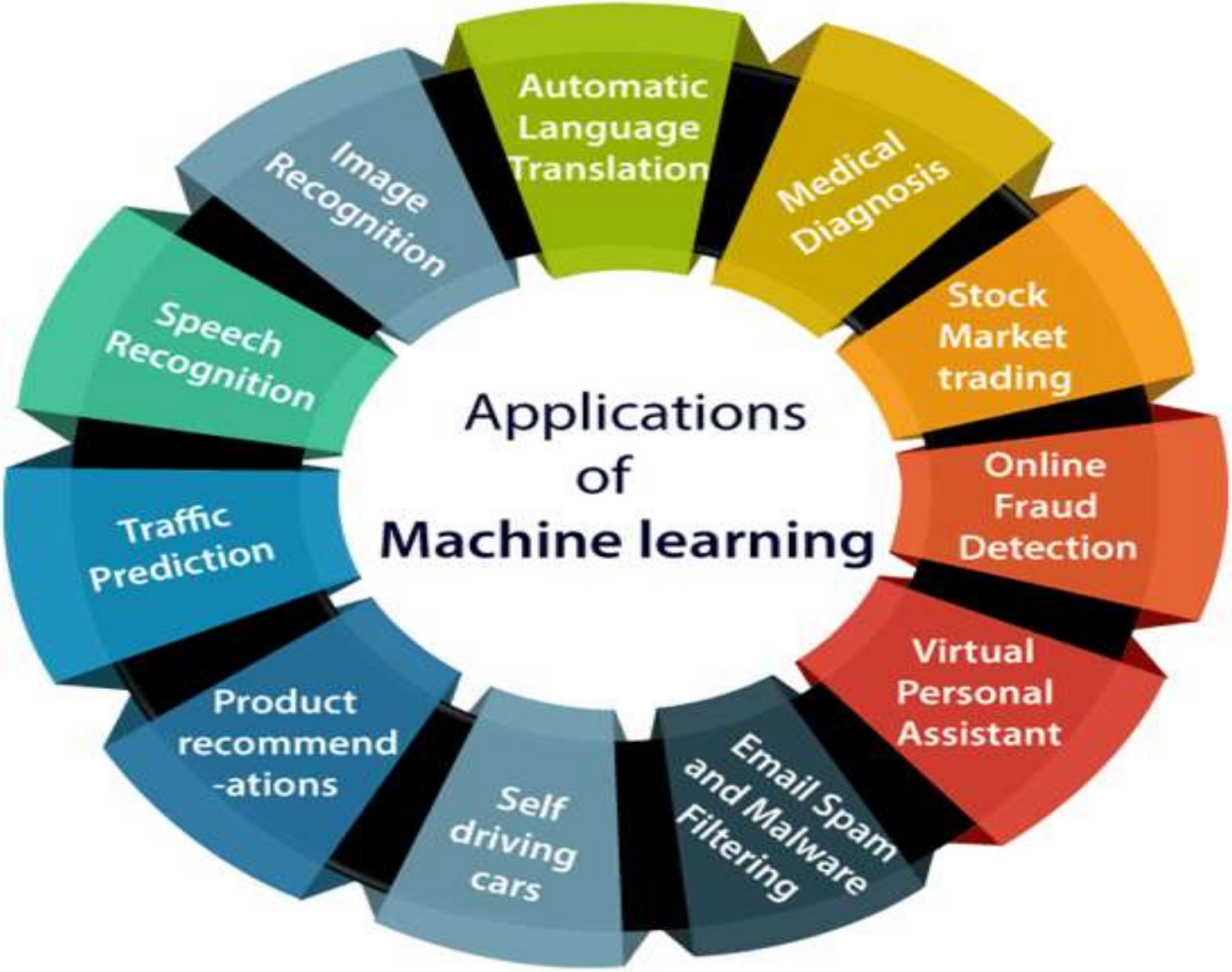
- **Model-Based:**
- In this Reinforcement Learning method, you need to create a virtual model for each environment. The agent learns to perform in that specific environment.

Reinforcement Learning	Supervised Learning
RL works by interacting with the environment.	Supervised learning works on the existing dataset.
The RL algorithm works like the human brain works when making some decisions.	Supervised Learning works as when a human learns things in the supervision of a guide.
There is no labeled dataset is present	The labeled dataset is present.
No previous training is provided to the learning agent.	Training is provided to the algorithm so that it can predict the output.
RL helps to take decisions sequentially.	In Supervised learning, decisions are made when input is given.


Machine Learning
Unsupervised Learning
Supervised Learning
Reinforcement Learning
Dimensionality Reduction
Clustering
Classification
Regression
Real-time Decisions
Robot Navigation
Learning Tasks
Skills Acquisition
Game AI
Big Data Visualization
Meaningful Compression
Structure Discovery
Feature Elicitation
Recommender Systems
Targetted Marketing
Customer Segmentation
Idenity Fraud Detection
Image Classification
Customer Retention
Diagnostics
Advertising Popularity Prediction
Weather Forecasting
Market Forecasting
Estimating Life Expectancy
Population Growth Prediction

Artificial Intelligence	Machine learning
Artificial intelligence is a technology which enables a machine to simulate human behavior.	Machine learning is a subset of AI which allows a machine to automatically learn from past data without programming explicitly.
The goal of AI is to make a smart computer system like humans to solve complex problems.	The goal of ML is to allow machines to learn from data so that they can give accurate output.
In AI, we make intelligent systems to perform any task like a human.	In ML, we teach machines with data to perform a particular task and give an accurate result.
Machine learning and deep learning are the two main subsets of AI.	Deep learning is a main subset of machine learning.
AI has a very wide range of scope.	Machine learning has a limited scope.
AI is working to create an intelligent system which can perform various complex tasks.	Machine learning is working to create machines that can perform only those specific tasks for which they are trained.
AI system is concerned about maximizing the chances of success.	Machine learning is mainly concerned about accuracy and patterns.
The main applications of AI are **Siri, customer support using catboats**, Expert System, Online game playing, intelligent humanoid robot, etc.	The main applications of machine learning are **Online recommender system**, **Google search algorithms**, **Facebook auto friend tagging suggestions**, etc.
On the basis of capabilities, AI can be divided into three types, which are, **Weak AI**, **General AI**, and **Strong AI**.	Machine learning can also be divided into mainly three types that are **Supervised learning**, **Unsupervised learning**, and **Reinforcement learning**.
It includes learning, reasoning, and self-correction.	It includes learning and self-correction when introduced with new data.
AI completely deals with Structured, semi-structured, and unstructured data.	Machine learning deals with Structured and semi-structured data.

Applications of machine learning



1. Image Recognition

- Image recognition is one of the most common applications of machine learning.

- It is used to identify objects, persons, places, digital images, etc. The popular use case of image recognition and face detection is, **Automatic friend tagging suggestion**:

- Facebook provides us a feature of auto friend tagging suggestion. Whenever we upload a photo with our Facebook friends, then we automatically get a tagging suggestion with name, and the technology behind this is machine learning's **face detection** and **recognition algorithm**.

2. Speech Recognition

- While using Google, we get an option of "**Search by voice**," it comes under speech recognition, and it's a popular application of machine learning.

- Speech recognition is a process of converting voice instructions into text, and it is also known as "**Speech to text**", or "**Computer speech recognition**."

- At present, machine learning algorithms are widely used by various applications of speech recognition.

- **Google assistant**, **Siri**, **Cortana**, and **Alexa** are using speech recognition technology to follow the voice instructions.

3. Traffic prediction

- If we want to visit a new place, we take help of Google Maps, which shows us the correct path with the shortest route and predicts the traffic conditions.

- It predicts the traffic conditions such as whether traffic is cleared, slow-moving, or heavily congested with the help of two ways:
  - **Real Time location** of the vehicle form Google Map app and sensors
  - **Average time has taken** on past days at the same time.

- Everyone who is using Google Map is helping this app to make it better.

4. Product recommendations:

- Machine learning is widely used by various e-commerce and entertainment companies such as **Amazon**, **Netflix**, etc., for product recommendation to the user.

- Whenever we search for some product on Amazon, then we started getting an advertisement for the same product while internet surfing on the same browser and this is because of machine learning.

5. Self-driving cars

- One of the most exciting applications of machine learning is self-driving cars. Machine learning plays a significant role in self-driving cars.

- **Tesla**, the most popular car manufacturing company is working on self-driving car.

- It is using unsupervised learning method to train the car models to detect people and objects while driving.

6. Email Spam and Malware Filtering

- Whenever we receive a new email, it is filtered automatically as important, normal, and spam.

- We always receive an important mail in our inbox with the important symbol and spam emails in our spam box, and the technology behind this is Machine learning.

- Below are some spam filters used by Gmail:
  - Content Filter
  - Header filter
  - General blacklists filter
  - Rules-based filters
  - Permission filters

- Some machine learning algorithms such as **Multi-Layer Perceptron**, **Decision tree**, and **Naïve Bayes classifier** are used for email spam filtering and malware detection.

7. Virtual Personal Assistant

- We have various virtual personal assistants such as **Google assistant**, **Alexa**, **Cortana**, **Siri**.

- As the name suggests, they help us in finding the information using our voice instruction.

- These assistants can help us in various ways just by our voice instructions such as Play music, call someone, Open an email, Scheduling an appointment, etc.

8. Online Fraud Detection

- Machine learning is making our online transaction safe and secure by detecting fraud transaction.

- Whenever we perform some online transaction, there may be various ways that a fraudulent transaction can take place such as **fake accounts**, **fake ids**, and **steal money** in the middle of a transaction.

- So to detect this, **Feed Forward Neural network** helps us by checking whether it is a genuine transaction or a fraud transaction.

- **9. Stock Market trading:**
- Machine learning is widely used in stock market trading. In the stock market, there is always a risk of up and downs in shares, so for this machine learning's **long short term memory neural network** is used for the prediction of stock market trends.

- **10. Medical Diagnosis:**
- In medical science, machine learning is used for diseases diagnoses. With this, medical technology is growing very fast and able to build 3D models that can predict the exact position of lesions in the brain.
- It helps in finding brain tumors and other brain-related diseases easily.

11. Automatic Language Translation

- Nowadays, if we visit a new place and we are not aware of the language then it is not a problem at all, as for this also machine learning helps us by converting the text into our known languages.

- Google's GNMT **(Google Neural Machine Translation)** provide this feature, which is a Neural Machine Learning that translates the text into our familiar language, and it called as automatic translation.

Tools for machine learning

- **<u>Python</u>**
- Python is one of the most popular programming languages of recent times.
- Python, created by **Guido van Rossum in 1991, is an open-source, high-level, general-purpose programming language.**
- Python is a dynamic programming language that **supports object-oriented, imperative, functional, and procedural development paradigms.**
- Python is very popular in machine learning programming.
- Python is one of the first programming languages that got the support of machine learning via a variety of libraries and tools.
- Scikit and TensorFlow are two popular machine learning libraries available to Python developers.

R

- R language is a **dynamic, array-based, object-oriented, imperative, functional, procedural, and reflective computer programming language.**
- The language first appeared in 1993 but has become popular in past few years among data scientists and machine learning **developers for its functional and statistical algorithm features.**
- R language was created by **Ross Ihaka and Robert Gentleman** at the University of Auckland, New Zealand.
- R is open-source and available on r-project.org and Github.
- Currently R is managed and developed under the R Foundation and the R Development Core Team.
- The current version of R is 3.5.2 that was released on Dec 20, 2018.
- R language is one of the **most popular programming languages among data scientists and statistical engineers.**
- R supports Linux, OS X, and Windows operating systems.

Matlab

- Matlab (Matrix laboratory) is a **licensed commericial software with a robust support for a wide range of numerical computing**.
- MATLAB has a huge user base across industry and academia.
- **MATLAB is developed by MathWorks.**
- MATLAB also provides extensive **support of statistical functions and has a huge number of machine learning algorithms in-built.**
- It also has the ability **to scale up for large dataset by parallel processing on cluster and cloud.**

SAS

- SAS **(earlier known as 'statistical Analysis System'**) is another licensed **commercial software which provides strong support for machine learning functionalities.**

- **Developed in C by SAS had its first release in the year 1976.**

- SAS is a **software suite computing different components**.

- The basic data management functionalities are embedded in the Base SAS component whereas the other components like SAS/INSIGHT, Enterprise Miner, SAS/STAT, etc. help in specialized functions related to data mining and statistical analysis.

Other languages/tools

- Owned by IBM, **SPSS** (**originally named as Statistical package for the social sciences**) is a popular package **supporting specialized data mining and statistical analysis.**

- **Julia is an open source**, liberal **license programming language for numerical analysis and computational science** and also having ability to implement high-performance machine learning algorithms.